The Resilient, but Limited Effect of Process on Fairness and Legitimacy: Evidence from Wind Energy

Michael Hankinson*

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Abstract

Renewable energy infrastructure often faces local opposition due to its spatially concentrated costs. However, a permitting process based on local control and high public input is believed to increase the fairness and legitimacy of even unpopular policy decisions. Yet most evidence relies on retrospective evaluations that may reflect whether individuals obtained their preferred siting outcome. Using two preregistered survey experiments, I isolate the causal effect of process from outcomes on the fairness and legitimacy of wind energy siting. The experiments independently vary who makes the siting decision, the level of public input, and whether respondents learn the outcome before evaluating the process. Centralized authority and limited public input reduce procedural fairness and legitimacy when outcomes are unknown, with public input having the larger effect. These effects persist even when respondents know whether the project has been approved or denied. However, these process effects are substantially smaller than that of getting one's preferred outcome, indicating that ex post complaints about the process are heavily shaped by frustration over the outcome. These comparatively limited effects of process should encourage the use of centralizing reforms which remove local veto points. Such reforms are unlikely to lose many would-be supporters on the basis of procedural fairness and can be paired with high levels of public input to maintain legitimacy without the local veto points.

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*Assistant Professor, Department of Political Science, George Washington University.

From wind turbines to transmission lines, renewable energy infrastructure has diffuse benefits and spatially concentrated costs, ranging from concerns over aesthetic impacts to the preservation of local identity (Bergquist et al. 2020; Devine-Wright and Howes 2010; Susskind et al. 2022; Wolsink 2007). Consequently, opponents are often able to leverage local permitting powers to block the construction of energy infrastructure critical for responding to climate change. To avoid this local bottleneck, governments across North America and Europe are increasingly moving permitting authority over renewable energy to centralized institutions and unelected expert bodies (Chaudhuri and Susskind 2024), insulating the decisions from public input and local accountability. Such reforms bring risk. If voters believe that the loss of local control and public input is unfair, they may be less willing to accept energy siting outcomes as legitimate. This centralization of authority risks degrading trust in government, sparking local protests, and exacerbating the right-wing populist backlash already associated with decarbonization (Bosetti et al. 2025).

Concern over centralization and decreased public input stems from the belief that procedural fairness is critical for sustaining policy legitimacy (Tyler 1994). When the public believes that decision makers are accountable or that the public has had meaningful opportunities for input, they may be more willing to accept policy outcomes, even ones they oppose. Applied to renewable energy, a large literature argues that residents are more supportive of project proposals that emerge from intense community engagement (e.g., Elmallah and Rand 2022; Firestone et al. 2018, 2020, 2012; Gadzanku et al. 2025; Gross 2007; Hoen et al. 2019; Hoesch et al. 2025). Despite this evidence, it remains unclear whether residents' perceptions of fairness independently shape legitimacy or whether evaluations of process simply reflect whether people secured their preferred outcome.

Identifying the causal effect of process on fairness and legitimacy is challenging for two reasons. First, most observational evidence is retrospective, measuring perceptions of fairness after a decision is known. Consequently, it is difficult to separate principled judgments about institutions from reactions to winning or losing. Second, changes in process can also affect

the outcome. For example, high levels of public input can lead to a more locally acceptable project. Thus, experimental designs which randomly vary the process but do not specify the outcome cannot isolate the effect of process on legitimacy (e.g., Liu et al. 2020; Wicki, Hofer and Kaufmann 2022).

The stakes are high. If the public's assessment of procedural fairness and legitimacy are driven primarily by outcomes, then institutions that favor local control and high public input may not only facilitate obstruction, but also do little to increase the acceptance of the renewable energy infrastructure that is sited. In contrast, if process does have an independent effect on fairness and legitimacy, then the centralization of authority risks eroding legitimacy and trust in government, threatening the long-term success of climate action.

Wind energy siting provides a ideal context for evaluating whether institutional design or outcomes have a greater effect on policy legitimacy. Survey evidence finds that a third of recent wind energy projects have been canceled due to community protest (Nilson et al. 2024) and the rate of local protest is growing (Stokes et al. 2023). Even in Denmark, a country with majority public support for expansion of wind energy, opposition pushes local officials to site turbines in the least politically damaging areas (Andersen et al. 2025). In response, many governments across North America and Europe have centralized siting authority to the state, regional, or even national level and limited requirements for public hearings in order to accelerate deployment (Chaudhuri and Susskind 2024; Stokes 2016; Uibeleisen and Götz 2024; Vasstrøm et al. 2024).

In this study, I isolate the effect of the decision-making process from outcomes using two preregistered survey experiments. Respondents from suburban and rural areas in the U.S. evaluate a proposed wind energy project under randomly assigned institutional arrangements that vary (i) who makes the decision—locally elected officials or a state-appointed board of experts—and (ii) the level of public input—face-to-face meetings or written comments only. Crucially, the experiments also randomize whether respondents learn the outcome before evaluating procedural fairness and legitimacy.

Across both experiments, three findings emerge. First, centralized decision making and limited public input reduce procedural fairness and the legitimacy of siting decisions when outcomes are unknown, with public input having the larger effect. Second, these effects are resilient to outcome information: learning that a proposal is approved or denied does not diminish the independent effect of process. Third, although procedural fairness matters, the effects of process are substantially smaller than the effect of receiving one's preferred outcome.

Together, these results clarify both the influence and the limits of procedural fairness in democratic governance. Institutional design can increase both the fairness of the siting process and the legitimacy of siting outcomes, even ones voters disagree with. But the effects of such processes only go so far. Fairness and legitimacy are more greatly influenced by whether voters agree with the outcome, meaning residents' ex post complaints about a fair process are heavily shaped by frustration over the outcome.

For policymakers, these comparatively small effects of process should encourage the centralization of energy siting. Such reforms are unlikely to lose many would-be supporters on the basis of procedural fairness. Combination reforms, such as centralized decision making with high public input, should minimize the loss of legitimacy while also removing local veto points. Of course, even small effects of process on legitimacy may be politically consequential if they are concentrated among a highly engaged, well-resourced faction willing to mobilize against the infrastructure. By isolating the effect of process from outcome, this study provides a framework for identifying such behavioral consequences in future research.

Main

Measuring the Effect of Process in Wind Turbine Siting

The effect of process independent of outcome can be viewed through *procedural fairness theory.* The three-step theory begins with a decision being made using a procedure, such

as the decision to build a piece of energy infrastructure. Next, an observer evaluates the fairness of the procedure. Finally, the observer's assessment of fairness combined with their predisposition towards the infrastructure leads to either the acceptance or rejection of the decision—otherwise known as legitimacy (Tyler 1994).

For observational research, separating the effect of process from the effect of the outcome is challenging. Voters may express attitudes about process, but abandon these procedural principles in favor of a preferred outcome (Prothro and Grigg 1960). As a result, it is reasonable to be skeptical when opponents claim that their concerns are over process and independent of the outcome.

Despite this challenge, isolating these independent forces is critical given the evolving context of energy siting. Since the 1970's, community engagement has been an essential part of infrastructure permitting. While most projects address regional needs, new infrastructure often has spatially concentrated costs, mobilizing local opposition. In response, the primary recommendation for winning local support has been greater local control and public input (e.g., Elmallah and Rand 2022; Firestone et al. 2018, 2020, 2012; Gadzanku et al. 2025; Gross 2007; Hoen et al. 2019; Hoesch et al. 2025). More community voice is believed to not only produce projects better tailored to local conditions, but also increase the legitimacy of even unpopular projects. But if appeals to procedural justice are just a tactic to secure one's preferred policy, then reforms which create more local veto points will not only constrain the supply of collective goods, but do nothing to increase voters' acceptance of unpopular outcomes.

Approaches to measuring the effect of process on fairness in energy siting largely rely on observational methods. Case studies and surveys build on retrospective correlations between the community engagement process and a sense of procedural fairness (Firestone et al. 2018, 2020; Gross 2007; Hoen et al. 2019; Hoesch et al. 2025) as well as support for the project overall (Nilson and Stedman 2023). For studies which only sample residents living near approved projects, researchers cannot observe projects which had processes deemed unfair

enough that opponents derailed approval. Furthermore, retrospective reflections struggle to account for endogeneity. Recalled preferences are not only subject to measurement error, but also Tiebout sorting. A wind farm's greatest opponents may be more likely to leave the area once siting is approved, whereas some supporters now living in the area were not present for the siting conflict.

Experimental designs can break this endogeneity (e.g., Liu et al. 2020; Wicki, Hofer and Kaufmann 2022). Though as stated, the design needs to specify and separately randomize both the process and the outcome. Otherwise, if respondents believe that more community engagement would lead to a "better" outcome—such as the project being denied—then even an experiment cannot disentangle process from outcome. Also, these designs often use a bundled treatments of community engagement without isolating the individual elements of the process, such the level of decision making versus public input.

Wind Turbine Siting Institutions

Isolating the relationship between institutions and procedural fairness is highly relevant for energy siting policy globally. Local governments typically control decision-making over new wind facilities, presenting obstacles to development even in countries where wind energy investment is popular, such as Denmark and Sweden (Andersen et al. 2025; Lindvall 2023; Vasstrøm et al. 2024). Compared to centralized authorities, local legislators are elected and thus incentivized to keep their votes more closely aligned with nearby residents' attitudes. Additionally, local legislators will typically hold more in-person public hearings on wind proposals, facilitating a higher level of face-to-face public input.

In contrast to this local control with high public input, governments across North America and Europe have sought to centralize decision-making authority. Recently, Germany curtailed local control by setting binding top-down wind energy targets and allowing the state and municipal authorities to manage planning and implementation. Additional reforms have removed public hearings for wind proposals unless requested by the developer or local au-

thorities (Uibeleisen and Götz 2024). The United Kingdom has similarly exempted large scale wind power installations from local planning approval. In contrast, Norway replaced local control over wind energy development with a national framework in 2009, sparking widespread protest. By 2019, Norway began a series of reforms to strength local control over wind siting, illustrating how a loss of legitimacy may threaten long-term policy success (Vasstrøm et al. 2024).

Rather than centralizing authority nationally, regional governments can also preempt local control. Canadian provinces have removed local veto authority, leading to backlash and the punishment of incumbent officials (Stokes 2016). In the U.S., states have also shifted decision-making over utility-scale wind and solar facilities from local governments to state-appointed boards of experts, including Illinois, California, New York, Wisconsin, and Michigan. Other states have had strong state control for much longer, including Connecticut, Maryland, West Virginia, and North Carolina. Another 20 states maintain a hybrid approach, where large-scale energy facilities and transmission lines are approved at the state level (Chaudhuri and Susskind 2024). While this preemption often requires at least one public hearing, the level of input typically falls short of that gathered through locally elected decision-making bodies.

Analytical Approach

Prior to data collection, I preregistered two experimental designs. The surveys were fielded on samples of suburban and rural respondents in the United States. For clarity, I describe the common points of inquiry below. Complete preregistrations and all preregistered tests are included in the Supplementary Information.

The experiments proposed a hypothetical wind farm within eyesight of a respondent's neighborhood, highlighting the scale of the proposal as well as expected local benefits and costs commonly discussed in siting campaigns. The proposal randomly varied the level of government making the decision, the level of public input, the outcome, and whether

respondents saw this outcome prior to evaluating the process. Dependent variables included the fairness of the decision-making process measured using a 7-item Likert scale and the legitimacy of the outcome. Legitimacy was measured using a novel index of four items designed to measure respondents' acceptance of the decision.

Figure 1 displays a stylized survey flow common across both experiments. The key points of inquiry are as follows:

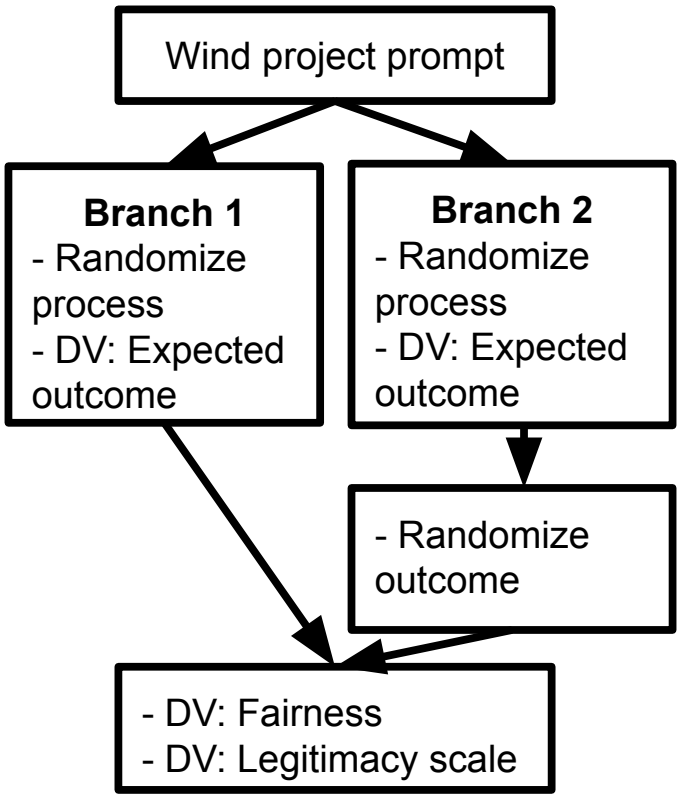


Figure 1: Stylized survey flow from both experiments.

What is the effect of state control and limited public input when the outcome is not known? To test the effect of state control and limited public input, I regress the fairness and legitimacy dependent variables on the process treatments using respondents who were not informed of the outcome (Branch 1), allowing for both of the above mechanisms.

Does knowing the outcome alter the effect of the process? If process has a sincere, independent effect on the fairness and legitimacy of decisions, then providing respondents with the outcome should not change the effect of process. In contrast, if outcomes drive evaluations of procedural fairness, then the effect of state control and limited input on fairness and legitimacy should be substantively smaller—if not null—for respondents who are informed of the outcome (Branch 2). To assess this relationship, I regress the fairness and legitimacy dependent variables on the process treatments interacted with the respondents' assigned branch using the full sample.

Effect of Process When Outcome Is Unknown

Figure 2 shows the effect of process on the fairness and legitimacy of the turbine siting among respondents who do not know the outcome (Branch 1 only). All results are reported from OLS regressions including demographic controls. See SI Section 9 for results in tabular form.

Experiment 1 uses a compound treatment of state-appointed board with limited public input in the form of written comments compared to locally elected officials with high public input in the form of face-to-face meetings. Respondents told that the decision would be made by a state-appointed board with limited public input were 10 percentage points ($d = 0.37$) less likely to perceive the decision-making process as fair. This effect is larger among those who tend to mostly agree with policies decisions made by their local government (SI Section 6). This 10 point effect is unchanged even when controlling for pretreatment support for the wind farm, concern for climate change, and the wind farm's expected effects on local property values (Table S6).

Experiment 2 decouples Experiment 1's compound treatment into separate treatments for the level of decision-making and public input. Having the decision made by a state-appointed board decreases fairness by 3 points ($d = 0.11$), compared to a decision by local elected officials. Holding no public meetings and only relying on written public comment decreases fairness by 9 points ($d = 0.29$), compared to holding in-person meetings. The

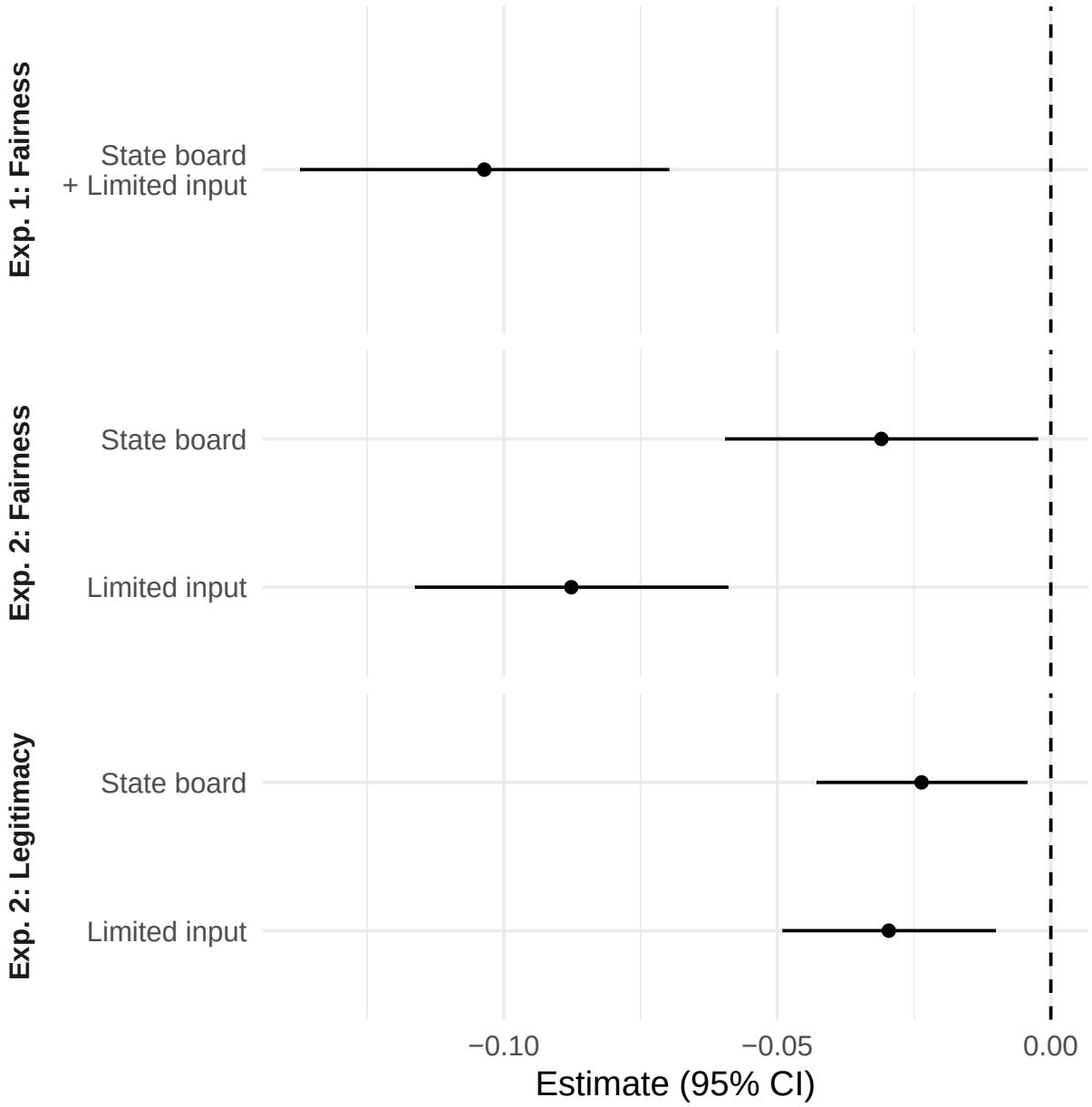


Figure 2: Effect of process on fairness and legitimacy of siting decision when outcome is unknown (Branch 1 only). Point estimates shown with 95% confidence intervals.

difference between these effects is statistically significant.

Experiment 2 also extends the effect of process to outcome legitimacy. Having the decision made by a state-appointed board of experts decreases legitimacy by 2 points ($d = 0.13$) and holding no public meetings decreases legitimacy by 3 points ($d = 0.15$). Altogether, Experiment 2 shows that the negative effects on fairness from Experiment 1 are driven primarily by the low level of public input, though both process treatments have comparable effects on outcome legitimacy. That the effect of process on legitimacy is smaller compared to that of fairness is expected. Legitimacy is a combination of one's baseline support for the outcome combined with their evaluation of procedural fairness (Tyler 1994).

Resilience of Effects When Outcome Is Known

Having established that process affects fairness and legitimacy absent knowledge of the outcome, I test whether people are similarly responsive to process when the outcome is known. Figure 3 shows the interaction between process and branch—whether the respondent knows the final decision. Across both experiments, knowing the outcome does not alter the effect of process on fairness and legitimacy. Whether or not the outcome is known, respondents still believe that the decision-making process is less fair and less legitimate when the decision is made by the state-appointed board or with limited public input. These null findings indicate that respondents are sincerely—not strategically—judging procedural fairness prior to knowing outcomes.

One concern may be that the null results in Figure 3 stem from respondents not internalizing the outcome. To test this theory, I interact the process treatments with an indicator for whether the outcome matches the respondent's pretreatment stated preference for the wind farm—either that it would be approved or denied. Figure 4 shows that receiving one's preferred outcome dramatically increases a respondent's perception of fairness and decision legitimacy by 11 to 16 percentage points, respectively. Respondents are internalizing the outcome and yet the outcome does not alter the independent effect of process. Nor does

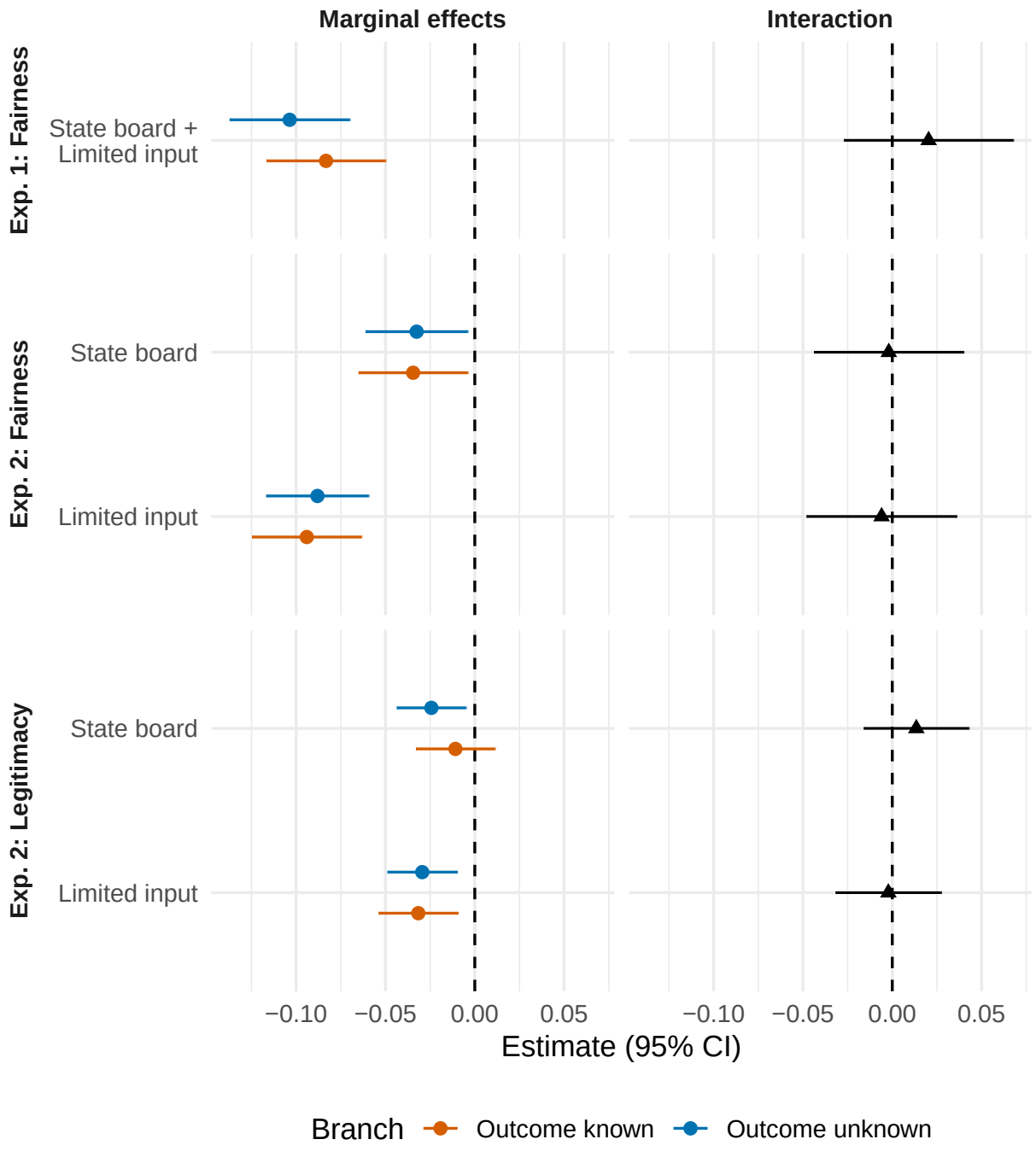


Figure 3: Interaction between knowledge of outcome and effect of process on fairness and legitimacy of siting decision (Branches 1 and 2). Point estimates shown with 95% confidence intervals.

getting one's preferred outcome alter the effect of process (SI Section 7).

However, Figure 4 also underscores the limited effects of process. These process effects are substantially smaller than the effect of getting one's preferred policy. Consequently, process may have little substantive effect on the acceptance of unpopular outcomes. Of course, other reforms—like early contact with nearby residents—may do more to legitimize policy decisions. Still, these findings suggest that studies which measure the effect of process after the outcome is known are capturing relationships mainly driven by whether the respondent got the outcome they wanted, not the independent effect of process alone.

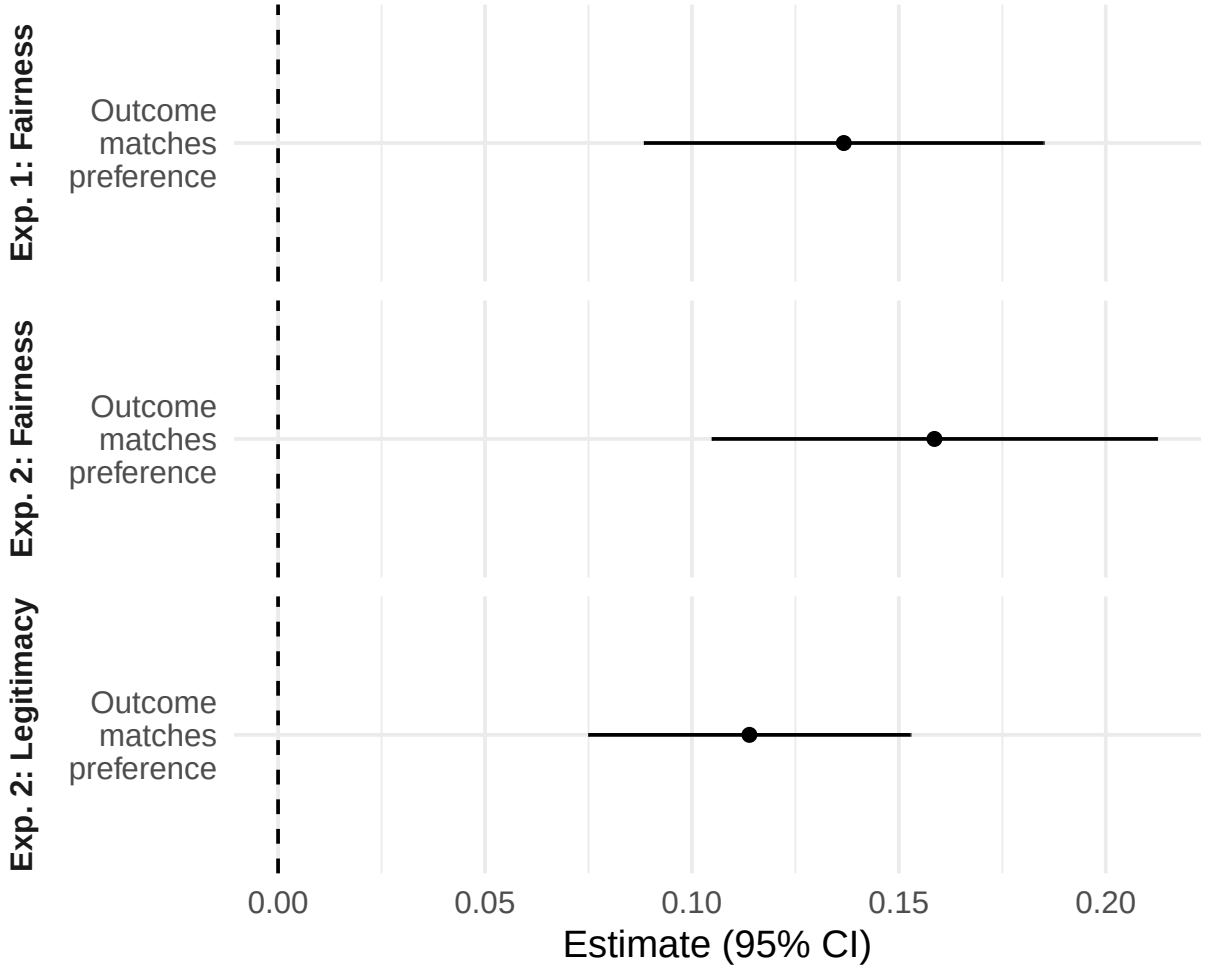


Figure 4: Effect of receiving preferred outcome on fairness and legitimacy of siting decision when outcome is known (Branch 2 only). Point estimates shown with 95% confidence intervals.

Discussion

Across two experiments, I demonstrate the causal effect of process on the fairness and legitimacy of wind energy siting. Face-to-face public input and, to a lesser extent, local control increase procedural fairness and legitimacy. These effects are resilient to knowing whether the institution will approve or deny the wind energy proposal. However, evaluations of process are even more heavily affected by the outcome, calling into question the substantive importance of procedural fairness in policy acceptance.

State and national governments across North American and Europe are increasingly preempting local control to facilitate the supply of new renewable energy. On one hand, these findings urge caution. This loss of local control and public input will make voters will be less accepting of the siting decisions. Over time, this depressed legitimacy risks degrading general trust in government and exacerbating right-wing populist backlash already associated with decarbonization (Bosetti et al. 2025). On the other hand, ex post evaluations of fairness are mainly driven by whether the voter received their preferred outcome, thus introducing bias in past efforts to study these risks.

Furthermore, the loss of legitimacy from centralization is relatively small compared to the ability of these reforms to expedite the deployment of renewable energy. Such reforms are unlikely to lose many would-be supporters due to procedural fairness. Instead, most project opponents would have opposed the siting outcome regardless of the process. Of course, the proposed benefits of community engagement are not limited to the approval of the project at hand, but also include managing broader intergroup conflict, maintaining trust in government, and building credibility in the promised economic benefits of the green transition (Beierle and Konisky 2000; Gazmararian and Tingley 2023). Thus, cautious policymakers may pair institutional features like face-to-face public input with centralized decision making in order to minimize a decline in legitimacy without creating local veto points.

Finally, these studies measure average treatment effects among the sample. If these small average effects are actually large effects concentrated in a subset of highly engaged, well-

resources opponents, then that faction may be influential enough to stymie the deployment of renewable energy. Future research should assess whether these relatively modest losses of legitimacy translate into behavioral backlash and meaningful consequences for renewable energy deployment. By isolating the effect of process from outcome, this study provides a framework for such an evaluation.

Methods

Sampling

Experiment 1 was fielded in December 2024. Respondents were recruited via Cloud Research Connect, an opt-in survey platform found to have among the highest response quality compared to similar popular platforms (Stagnaro et al. 2024). After dropping respondents for failing pretreatment attention checks, I surveyed 1,995 respondents living in the United States, age 18 years or older. To maximize external validity, I excluded respondents living in urban areas, as a wind farm of 340 foot tall turbines is an unrealistic land use in that context. As a result, the sample has a larger share of homeowners and non-Hispanic white respondents compared to the United States as a whole. See SI Section 3 for full descriptive statistics.

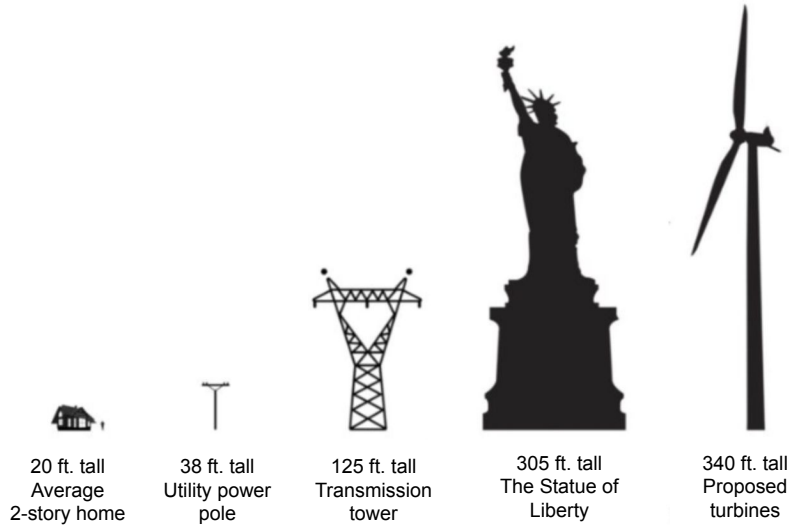
Experiment 2 was fielded in November 2025. I explicitly sampled respondents from rural areas in order to capture those most likely to encounter large-scale wind energy projects. Cloud Research Connect does not have a deep pool of rural respondents, so I fielded the survey via PureSpectrum, a platform which combines geographic micro-targeting with demographic quotas to recruit respondents from other opt-in survey providers. I used the University of Washington RUCA-to-ZCTA crosswalk, defining rural ZIP codes as those categorized with a RUCA code of 7 or higher. Demographic target quotas for income, race, homeownership, age, gender, and education are based on 2021-2025 5-year ACS estimates for these rural ZCTAs.

Both experiments were approved by the George Washington University and Princeton University Institutional Review Boards. Respondents were provided a consent form at the start of each survey and affirmatively expressed consent to advance in the study.

Design

In both survey experiments, respondents were asked to evaluate a hypothetical wind farm of sixty 340 foot tall turbines proposed for their community. There were explicitly told that this was not a real proposal, but to thoughtfully consider it as if it were real. The prompt includes both arguments for and against the wind farm, as well as a visualization of the scale of the turbines. Due to the diverse rural geographies in the United States, this study was unable to incorporate photomontages that would represent each respondents' surrounding context (e.g., woods v. grasslands, flat plains v. ridges, etc.). Below is the prompt used:

> A wind energy developer is proposing to build a wind farm on land in your community. They want to build a farm of 60 turbines, each around 340 feet tall, which would power 85,000 homes per year. [1] You would be able to see the wind turbines from your neighborhood.
>
> For context, here's a visual of the turbine height (far right) compared to other known structures:



[1] The energy production estimate was based on reporting of the 60 turbine Apex Clean Energy's Honey Creek project. However, the turbine height at Honey Creek is taller than the 340 feet described. In SI Section 3, I discuss the implications of this mismatch for my findings.

> Your community is divided. On one hand, some residents are concerned that the facility's size would harm local property values and affect migrating birds. Residents are also concerned about the aesthetics, landscape change, and potential negative impacts on quality of life due to the turbines.
>
> On the other hand, technical reports estimate that the wind farm would bring over $2 million per year in new tax revenue for local schools and services and reduce greenhouse gas emissions. Finally, others argue that farmers and landowners should have the right to lease their property to wind developers.

In both survey experiments, respondents were randomized into one of two branches. In Branch 1, respondents viewed only the process but they were not told the outcome of the policy siting decision. In Branch 2, respondents viewed both the process and the outcome. I also randomized the level of local control and public input. In Experiment 1, the "State board + limited input" condition presents a process where decisions are made by a state-appointed board of energy experts with limited, remote community input ("calling or writing an email to your governor"). The other condition reflects the status quo where decisions over wind turbines are made by elected local legislators with high levels of face-to-face community input ("attending local meetings and voicing your concerns face-to-face with your elected legislators").

In Experiment 2, I decouple this treatment into a level of government and level of public input treatments which vary independently of each other. For level of government, the conditions are:

- "your locally elected legislators"
- "a state-appointed board of energy experts"

For public input, the conditions are:

- "hold a series of local public meetings where residents like you can share your opinions face-to-face with the decision makers"

- "set up an email account where residents like you can send your thoughts, but there will be no public meetings about the proposal"

Finally, I randomized whether the proposed projected is approved or denied by its respective decision-making body. This randomized outcome was only visible for respondents in Branch 2, where the outcome is known. The survey flows for each experiment are diagrammed in Figures SI1 and SI3, as well as stylized survey flow in Figure 1.

The primary outcomes of interest are the fairness of the wind turbine siting process and the legitimacy of the final decision. For fairness, the survey asks: "Regardless of project approval, do you believe this is a fair way to decide on where to permit new wind energy farms?" Respondents indicated the fairness on a 7-point scale from "very unfair" to "very fair." I operationalize this outcome on a continuous scale from 0 to 1.

For legitimacy, I use an original four item scale of statement with agreement ranging on a 5-points scale from "strongly agree" to "strongly disagree." I average the four items. The four items are internally consistent with a Cronbach's alpha of .70. See SI Section 5 for validation of the legitimacy scale. The items are as follows, with the order randomized: "Thinking about the process and [eventual / NA] decision, do you agree or disagree with each of the following statements?"

- "This decision ought to be accepted and considered the final word on the matter."
- "The body that made this decision had the right to do so."
- "There ought to be an effort to challenge the decision and get it changed." *(reverse-coded)*
- "Even if I personally disagree, I respect the outcome of this process."

Statistical Analyses

All models used OLS regressions with robust standard errors.

$$y_i = \alpha + \beta D_i + \gamma X_i + \varepsilon_i$$

where y_i is either the fairness or legitimacy outcome for individual i; D_i is the process treatment for individual i; X_i is a $1 \times K$ vector of demographic covariates (e.g., homeownership, race, income, age, gender, education) with associated coefficients γ; and ε_i is an error term. All dependent variables are be scaled to a 0 to 1 outcome. I control for race/ethnicity, gender, education, age, income, homeownership, and partisanship. Education is defined as having completed college, partisanship ranges from -1 (Democrat/lean Democrat) to +1 (Republican/lean Republican), homeownership is a binary (homeowner or lives with homeowner), and race/ethnicity is operationalized as a binary for white, non-Hispanic due to the small share of non-white residents in the rural population of interest.

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Supplemental Information

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A Experiment 1 Survey Instrument

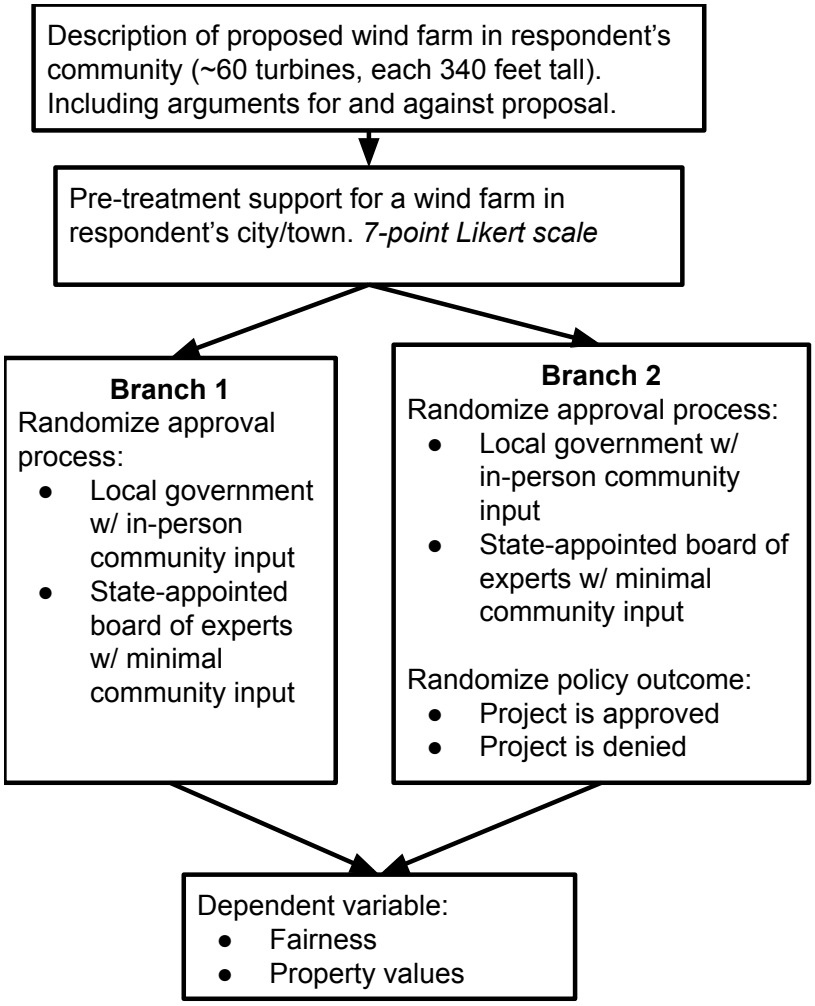

Figure A-1: Diagram of Experiment 1 survey flow.

Consent + Demographics

Standard module of demographic questions and attention checks.

From what you know about global climate change or global warming, which one of the following statements comes closest to your opinion?

- Global climate change has been established as a serious problem, and immediate action is necessary
- There is enough evidence that climate change is taking place and some action should be taken

- We don't know enough about global climate change, and more research is necessary before we take any actions
- Concern about global climate change is exaggerated. No action is necessary
- Global climate change is not occurring; this is not a real issue

In the United States, local governments are in charge of local public services, property taxes, and how land can be used. On a scale from 1 to 7, how often do you agree with the decisions made by your local government?

- Never
- Very rarely
- Rarely
- Sometimes
- Often
- Very often
- Always
- I don't know

Transition

We will now show you information regarding a potential scenario in your community. After reading the text, please give your opinions about the topic and other current events. Please pay close attention while reading, as you will be asked questions about the content of this scenario after you've read it.

Project Description

A wind energy developer is proposing to build a wind farm on land in your community. They want to build a farm of 60 turbines, each around 340 feet tall, which would power 85,000 homes per year. You would be able to see the wind turbines from your neighborhood.

For context, here's a visual of the turbine height (far right) compared to other known structures:

Your community is divided. On one hand, some residents are concerned that the facility's size would harm local property values and affect migrating birds. Residents are also concerned about the aesthetics, landscape change, and potential negative impacts on quality of life due to the turbines.

On the other hand, technical reports estimate that the wind farm would bring over $2 million per year in new tax revenue for local schools and services and reduce greenhouse gas emissions. Finally, others argue that farmers and landowners should have the right to lease their property to wind developers. Finally, others argue that farmers and landowners should have the right to lease their property to wind developers, regardless of the outcomes.

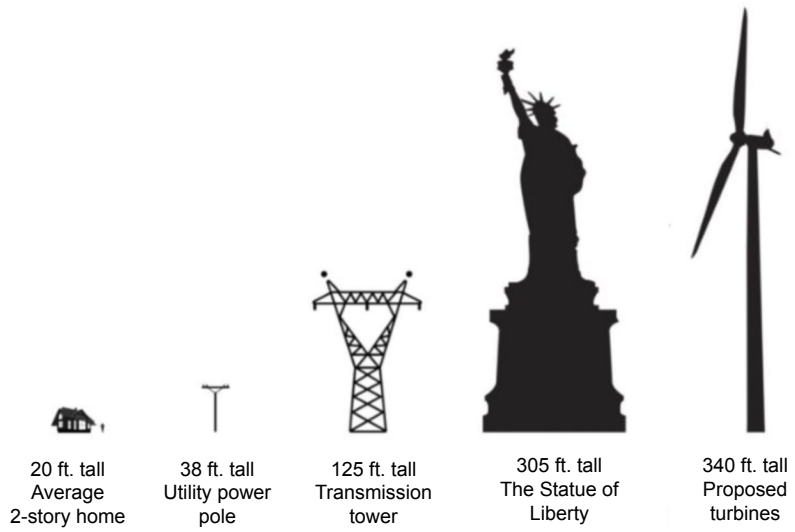

Figure A-2: Visualization of size of each turbine.

Pretreatment Support for Project

Do you support or oppose the development of this wind farm in your community?

- Strongly oppose
- Oppose
- Somewhat oppose
- Neither support nor oppose
- Somewhat support
- Support
- Strongly support

In at least two sentences, why do you support or oppose the wind farm? *Text box answer.*

Treatment 1: Process

According to state law, [your locally elected legislators gets / a state-appointed board of energy experts get] to decide on whether the proposed wind farm will be approved or denied. Local residents like yourself can share your opinions by [attending local meetings and voicing your concerns face-to-face with your elected legislators / calling or writing an email to your governor.]

Treatment 2: Policy Outcome (included in Branch 2 only)

After much debate, [your elected local legislators / the state-appointed board of experts] decided to [approve / reject] the wind farm proposal. It will [now / not] be built and this decision is final.

Dependent Variables (both Branch 1 and Branch 2)

Regardless of what you think about the [eventual / *NA*] decision, do you believe that a vote by [your elected local legislators/the state-appointed board of experts] is a fair way to decide on the approval of new wind energy farms?

- Very unfair
- Unfair
- Somewhat unfair
- Neither fair nor unfair
- Somewhat fair
- Fair
- Very fair

In at least two sentences, why do you believe this method was fair or unfair? *Text box answer.*

Placebo Check

In your opinion, how [would/will/would have] this proposed wind farm [affect/affected] property values in your community?

- 7 - Greatly increase property values
- 6
- 5
- 4 - No effect on property values
- 3
- 2
- 1 - Greatly decrease property values

Language conditionals:

- "would" - under Branch 1 - outcome not reported

- “will” - under Branch 2 - project approved
- “would have” - under Branch 2 - project rejected

In your opinion, which of the following processes would be the best way to decide on the approval of new wind farms?

- Majority vote by a state-appointed board of energy experts
- Majority vote by elected local government officials
- Majority vote by residents of the community
- Other: [text entry]
- I don’t know

B Experiment 2 Survey Instrument

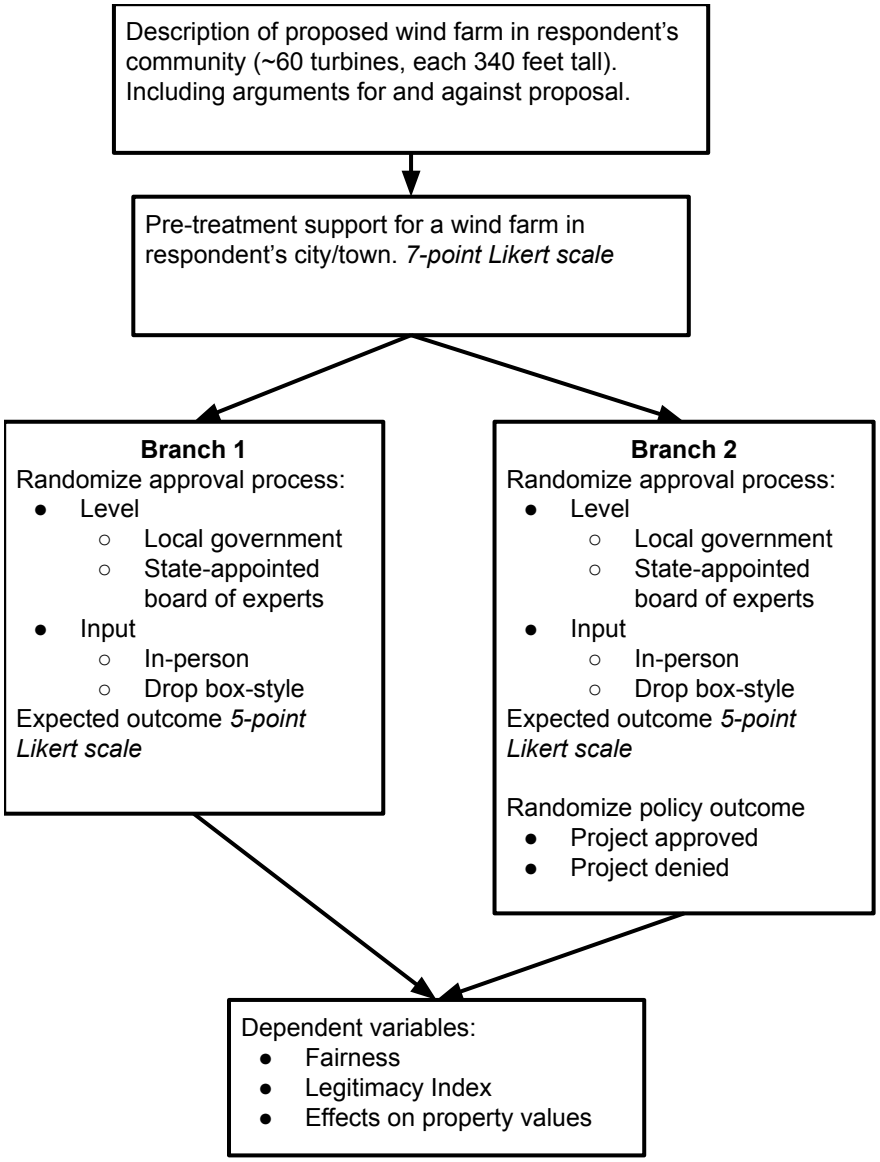

Figure B-3: Diagram of Experiment 2 survey flow.

Consent + Demographics

Standard module of demographic questions and attention checks.

From what you know about global climate change or global warming, which one of the following statements comes closest to your opinion?

- Global climate change has been established as a serious problem, and immediate action is necessary

- There is enough evidence that climate change is taking place and some action should be taken
- We don't know enough about global climate change, and more research is necessary before we take any actions
- Concern about global climate change is exaggerated. No action is necessary
- Global climate change is not occurring; this is not a real issue

In the United States, local governments are in charge of local public services, property taxes, and how land can be used. On a scale from 1 to 7, how often do you agree with the decisions made by your local government?

- Never
- Very rarely
- Rarely
- Sometimes
- Often
- Very often
- Always
- I don't know

Now, think about your state government. How often do you agree with the decisions made by your state government?

- Never
- Very rarely
- Rarely
- Sometimes
- Often
- Very often
- Always
- I don't know

Transition

We will now show you information regarding a potential scenario in your community. After reading the text, please give your opinions about the topic and other current events. Please pay close attention while reading, as you will be asked questions about the content of this scenario after you've read it.

Project Description

A wind energy developer is proposing to build a wind farm on land in your community. They want to build a farm of 60 turbines, each around 340 feet tall, which would power 85,000 homes per year. You would be able to see the wind turbines from your neighborhood.

For context, here's a visual of the turbine height (far right) compared to other known structures:

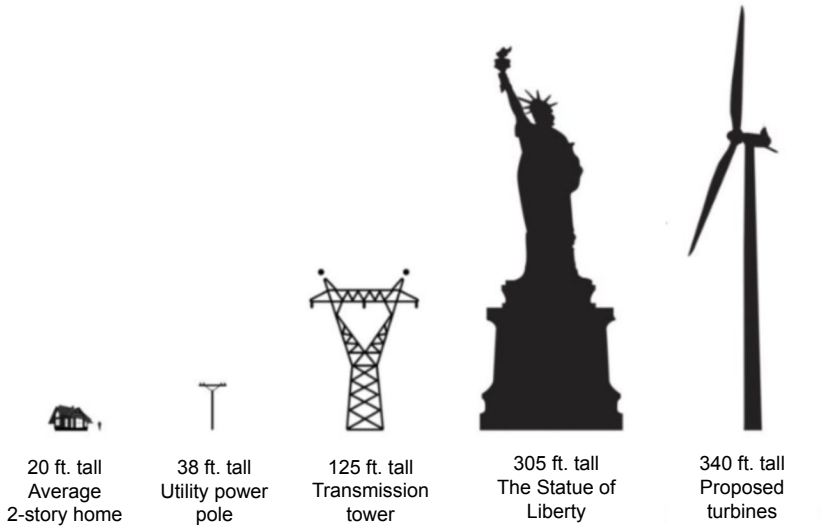

Figure B-4: Visualization of size of each turbine.

Your community is divided. On one hand, some residents are concerned that the facility's size would harm local property values and affect migrating birds. Residents are also concerned about the aesthetics, landscape change, and potential negative impacts on quality of life due to the turbines.

On the other hand, technical reports estimate that the wind farm would bring over $2 million per year in new tax revenue for local schools and services and reduce greenhouse gas emissions. Finally, others argue that farmers and landowners should have the right to lease their property to wind developers. Finally, others argue that farmers and landowners should have the right to lease their property to wind developers, regardless of the outcomes.

Pretreatment Support for Project

Do you support or oppose the development of this wind farm in your community?

- Strongly oppose
- Oppose
- Somewhat oppose
- Neither support nor oppose
- Somewhat support
- Support
- Strongly support

In at least two sentences, why do you support or oppose the wind farm? *Text box answer.*

Treatment 1: Process

According to state law, [your locally elected legislators gets / a state-appointed board of energy experts get] to decide on whether the proposed wind farm will be approved or denied. [Your locally elected legislators / A state-appointed board of energy experts] will [hold a series of local public meetings where residents like you can share your opinions face-to-face with the decision makers / set up an email account where residents like you can send your thoughts, but there will be no public meetings about the proposal].

Expected Outcomes

How likely do you think it is that the wind farm proposal will be approved?

- Extremely likely
- Likely
- Somewhat likely
- Equally likely and unlikely
- Somewhat unlikely
- Very unlikely
- Extremely unlikely

Treatment 2: Policy Outcome (included in Branch 2 only)

After much debate, [INSTITUTION] decided to [OUTCOME] the wind farm proposal. The wind farm will [MODIFIER] be built.

- *INSTITUTION: four unique conditions from fully randomizing the decision making body and level of public input.*
- *OUTCOME: deny / approve*
- *MODIFIER: not / now*

Dependent Variables (both Branch 1 and Branch 2)

Regardless of what you think about the [eventual / *NA*] decision, do you believe that a vote by [INSTITUTION]—after gathering input from residents through [email / public meetings]—is a fair way to decide on the approval of new wind energy farms?

- Extremely unfair
- Unfair
- Somewhat unfair
- Neither fair nor unfair
- Somewhat fair
- Fair
- Extremely fair

Thinking about the process and [eventual / NA] decision, do you agree or disagree with each of the following statements? Matrix table, 5-point scale (strongly agree, agree, neither agree nor disagree, disagree, strongly disagree)

- "This decision ought to be accepted and considered the final word on the matter."
- "People would be justified in protesting this decision." *(Reverse-coded). Note: This item was excluded from the analysis based on protocol specified in the pre-analysis plan. See Section E for discussion. Results are substantively and statistically unchanged regardless of its exclusion.*
- "The body that made this decision had the right to do so."
- "There ought to be an effort to challenge the decision and get it changed." *(Reverse-coded)*

- "Even if I personally disagree, I respect the outcome of this process."

In at least 2 sentences, why do you believe this [eventual / NA] decision [would be / is] either legitimate or illegitimate? *Text box answer*

Placebo Check

In your opinion, how [would/will/would have] this proposed wind farm [affect/affected] property values in your community?

- 7 - Greatly increase property values
- 6
- 5
- 4 - No effect on property values
- 3
- 2
- 1 - Greatly decrease property values

Language conditionals:

- "would" - under Branch 1 - outcome not reported
- "will" - under Branch 2 - project approved
- "would have" - under Branch 2 - project rejected

C Turbine Height Implications

Modern onshore wind turbines have an average hub height of 339 feet (Hartman 2024). Unfortunately, I interpreted that hub height as a tip of blade height in my visualization of the proposed turbines. How may this error have affected the experimental results? Taller turbines are likely to lead to more local opposition, but it is unclear if taller turbines would alter the legitimizing effect of the approval process. Instead, my expectation is that taller turbines would lead to an intercept shift downward—less support and less legitimacy overall—but the process conditions would have similar legitimizing treatment effects.

D Descriptive Statistics

Tables D-1 and D-2 compare each experiment's sample to its relevant population. Fielded on Cloud Research Connect, Experiment 1 sampled rural and suburban ZIP codes. To define the comparison group, I use the ZIP codes from which at least one respondent has entered Experiment 1's sample. For demographic traits, I use NHGIS to collect ZIP (ZCTA)-level data from the 2023 5-year American Community Survey (Schroeder et al. 2025). These ZIP code traits are weighted by their population to reflect the composition of residents themselves.

Experiment 1's sample reflects the population's composition of homeownership, gender, and age. Respondents in the sample have slightly lower incomes than the comparison population and are more likely to be white and college educated. The sample is 30% Republican or leans Republican. While partisanship data is not captured in the US Census, I believe that the ZIP population of these areas is slightly more Republican than the sample.

Experiment 2 samples exclusively rural ZIP codes and used demographic quotas based on the ZIP code data. While PureSpectrum has a deeper pool of rural respondents, these quotas were not strict. Still, the sample reflects the rural target population in terms of homeownership, college education, and income. The sample has an over-representation of women, white respondents, and older respondents. The sample is majority (54%) Republican/Republican leaners which is a reasonable target.

Although both samples exhibit modest deviations from their target populations on several demographic traits, these imbalances are unlikely to bias the estimated treatment effects. The treatments in both experiments are fully randomized at the individual level, and all observed demographic traits are controlled for in the analysis. As a result, differences between the sample and the target population may affect the levels of outcomes but not the internal validity of causal comparisons across treatment conditions (Coppock and McClellan 2019).

Table D-1: Descriptive statistics: Experiment 1 and weighted average of sampled ZIP codes

Characteristic	Survey sample	ZIP population (weighted)
Homeowner (pct.)	71%	68%
Female (pct.)	56%	51%
White, non-Hispanic (pct.)	74%	60%
Income (mean, household)	$85,814	$91,997
Age (mean)	42	39
College educated (pct.)	55%	39%
Republican/lean R. (pct.)	30%	NA

Table D-2: Descriptive statistics: Experiment 2 and weighted average of sampled ZIP codes

Characteristic	Survey sample	ZIP population (weighted)
Homeowner (pct.)	77%	72%
Female (pct.)	61%	50%
White, non-Hispanic (pct.)	85%	73%
Income (mean, household)	$68,293	$66,102
Age (mean)	50	42
College educated (pct.)	28%	24%
Republican/lean R. (pct.)	54%	NA

E Validation of Legitimacy Scale

The survey instrument for Experiment 2 included an original five item legitimacy scale. The items were as follows, with the order randomized: "Thinking about the process and [eventual / NA] decision, do you agree or disagree with each of the following statements?"

- "This decision ought to be accepted and considered the final word on the matter."
- "People would be justified in protesting this decision." *(Reverse-coded).*
- "The body that made this decision had the right to do so."
- "There ought to be an effort to challenge the decision and get it changed." *(Reverse-coded)*
- "Even if I personally disagree, I respect the outcome of this process."

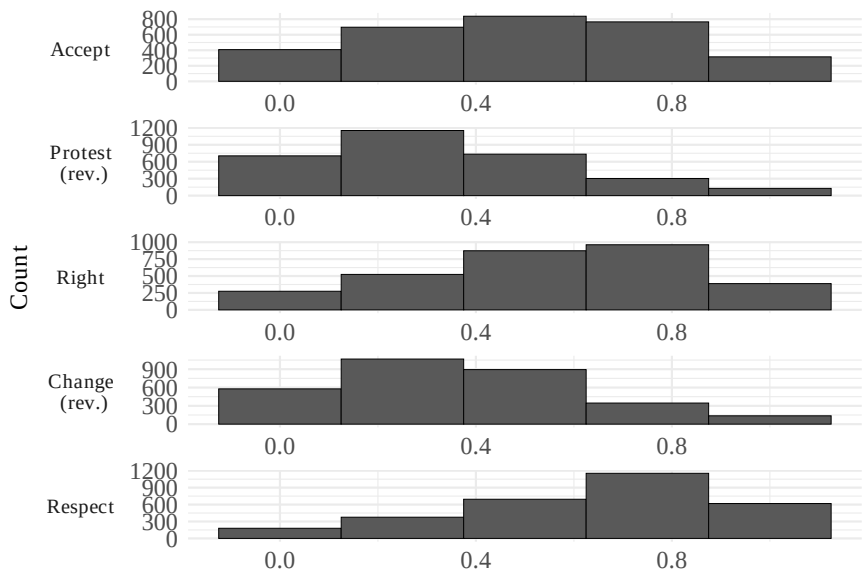

Figure E-5: Histograms of legitimacy scale items. Scaling for items "Protest" and "Change" already reverse-coded to match scale.

Table E-3: Scale reliability summary

Scale	Alpha_raw	Alpha_std	Avg_item_r	G6_smc	S_over_N
Legitimacy scale	0.69	0.69	0.31	0.69	2.24

Figure E-5 shows the distribution of support for each statement in the legitimacy scale. Support is reverse-coded for the "Protest" and "Change" items. While support for these two statements is lower than for the others, the heatmap shows positive correlations among

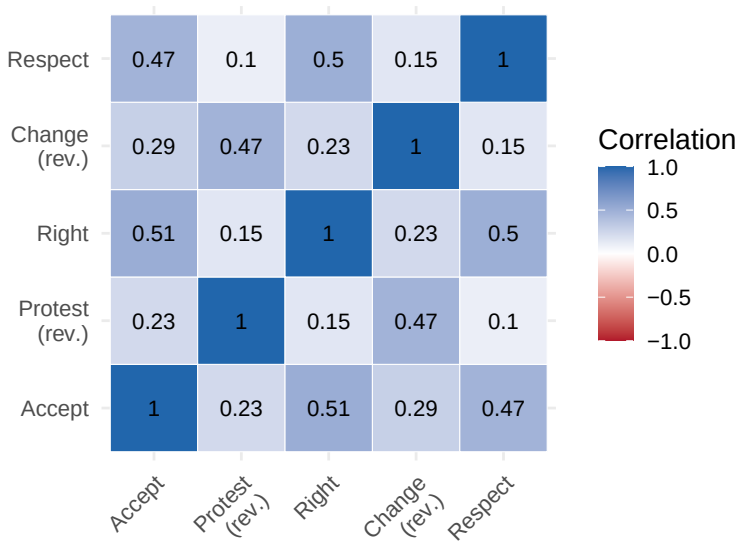

Figure E-6: Correlation heatmap of legitimacy scale items.

all five variables (Figure E-6). Finally, Table S-3 reports summary statistics on the study's novel legitimacy scale.

The pre-analysis plan notes that items would be dropped to target a Chronbach's alpha of .70 or higher. With all five items, the Chronbachs' alpha is .69. Dropping the "Protest" item increases the Chronbach's alpha to .70. Results are substantively and statistically the same in either approach (Figure E-7). However, to strictly adhere to the pre-analysis plan, the manuscript reports estimates using the four-item legitimacy, excluding the "Protest" item.

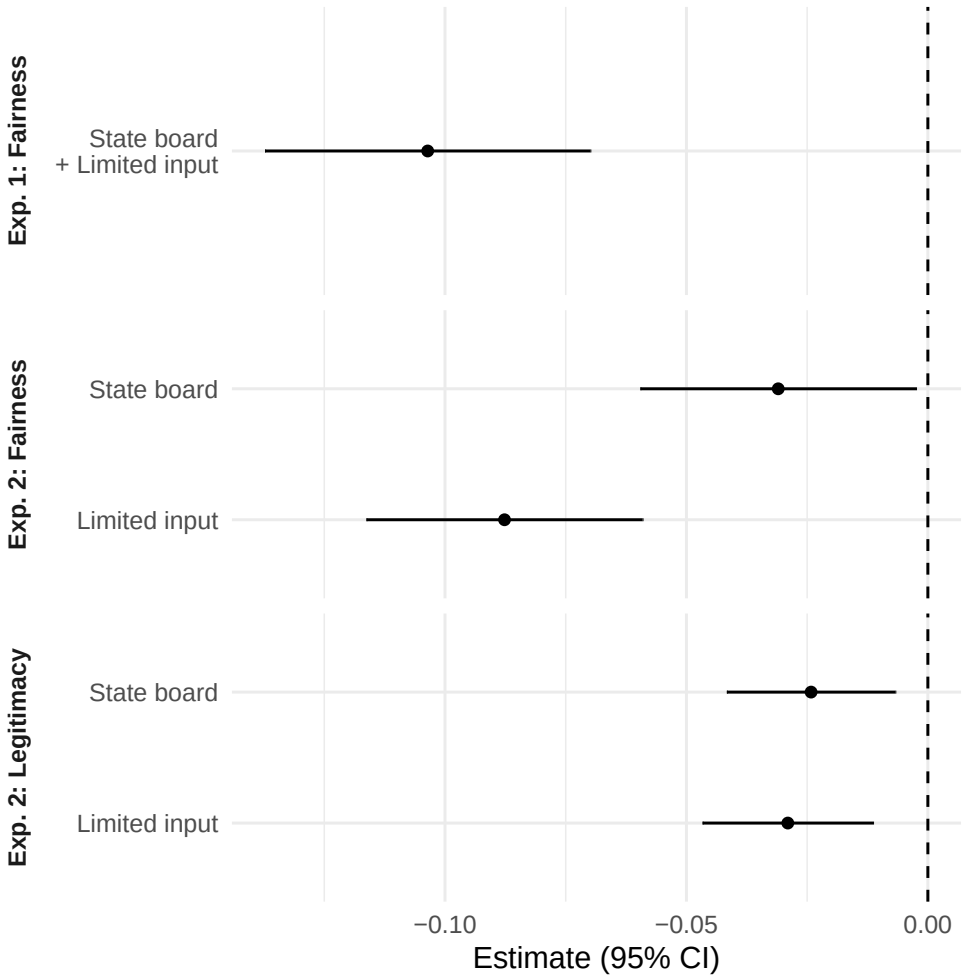

Figure E-7: Effect of process on perceived fairness and legitimacy of turbine siting decision when the outcome is unknown (Branch 1 only), using the original 5-item legitimacy scale. Compared to Figure 1. Point estimates shown with 95% confidence intervals.

F Alignment with Local Government

One explanation for the effect of level of decision making is that respondents may feel poorly represented by their state government. To test this mechanism, Experiment 1 asked: "How often do you agree with the decisions made by your local government?" I standardize this outcome as a two standard deviation effect to make it akin to a binary variable (Gelman 2008).

Of course, some respondents may feel poorly represented at the state and local level. Experiment 2 instead asked respondents to evaluate their agreement with decisions by their local government and state government. I subtract agreement with state government with agreement with local government to generate a cleaner measure of preference for local government control.

Figure F-8 shows the interactive effect of process with a preference for local government. In Experiment 1, for those who prefer local government, the compound treatment of state board with limited input suggests an even greater decrease in perceived fairness ($p = .07$), though this measurement of local government preference has flaws. In Experiment 2, for those who prefer local government, a decision made by state board of experts has a similar negative effect of fairness and legitimacy ($p < .05$). Of course, one's preference for local government is not randomly assigned, so these respondents may differ in other ways I cannot account for.

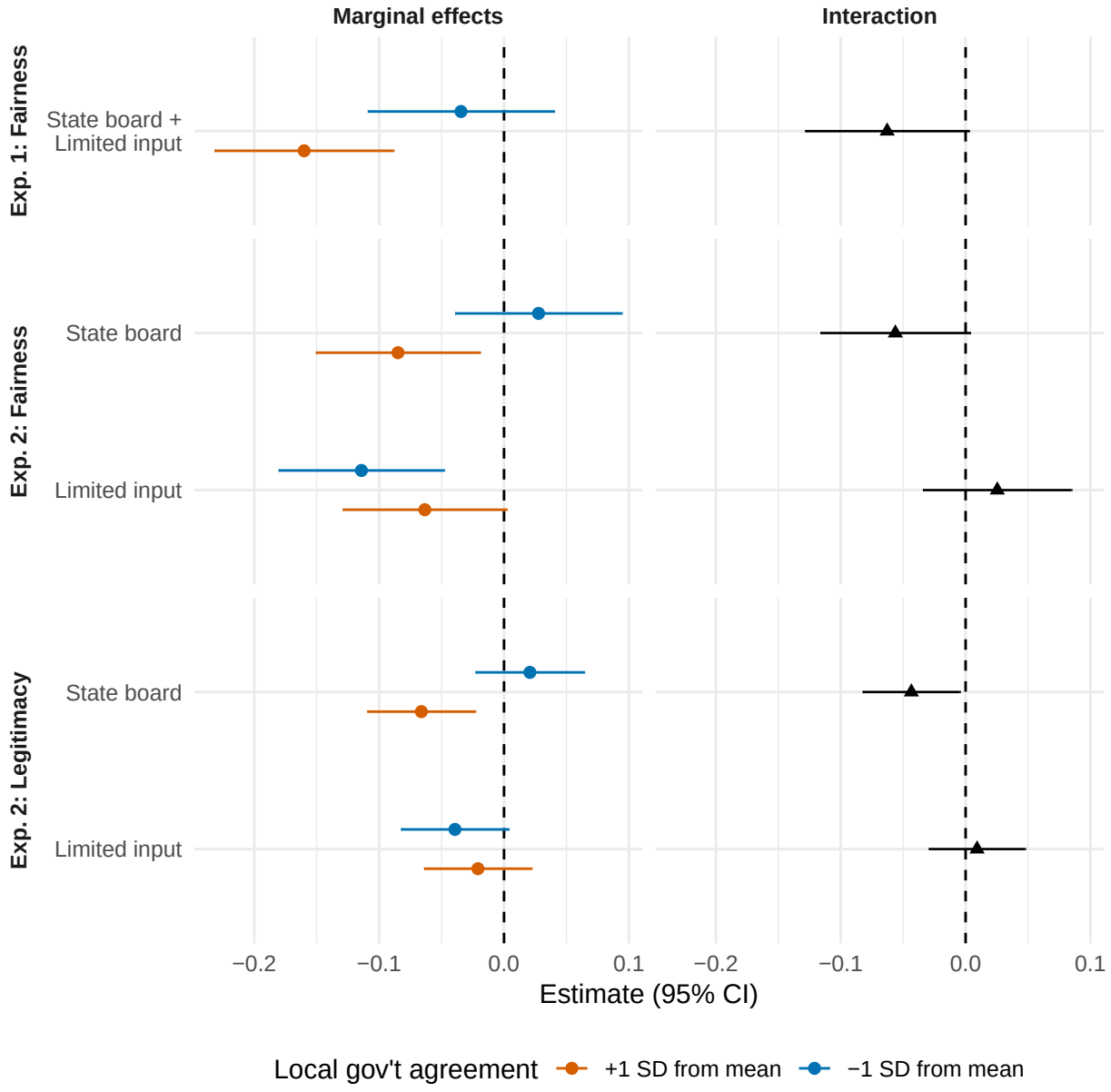

Figure F-8: Interaction between preference for local government and effect of process on perceived fairness and legitimacy of turbine siting decision when the outcome is unknown (Branch 1 only). Point estimates shown with 95% confidence intervals.

G Outcome Matches and Perceptions

Figure G-9 shows the interaction between match and the process. In 4 out of 5 treatments, getting one's preferred outcome does not alter the negative effects of process found in Figure 1. However, in Experiment 2, the interaction between outcome match and state board is large enough to negate the negative effect of state board. This suggests that people who receive their preferred outcome are no longer sensitive to who made the decision when judging the fairness of the process. However, those who *did not* receive their preferred outcome still perceive the state board as less fair than the local elected officials.

It is unclear why this may be the case and I hesitate to overanalyze one interaction from one of the treatments due to the risk of a false positive. Though, for theory generation, perhaps the level of decision maker is fungible in a way that the amount of public input is not. It may be easier to discount a state board as either corrupt or unimportant depending on the outcome compared to discounting public input. Public input may be seen as uniformly fair, regardless of the outcome. That this interaction exists in Experiment 2 but there is no evidence of it in Experiment 1 suggests that more data is needed to understand the stability and meaning of the relationship.

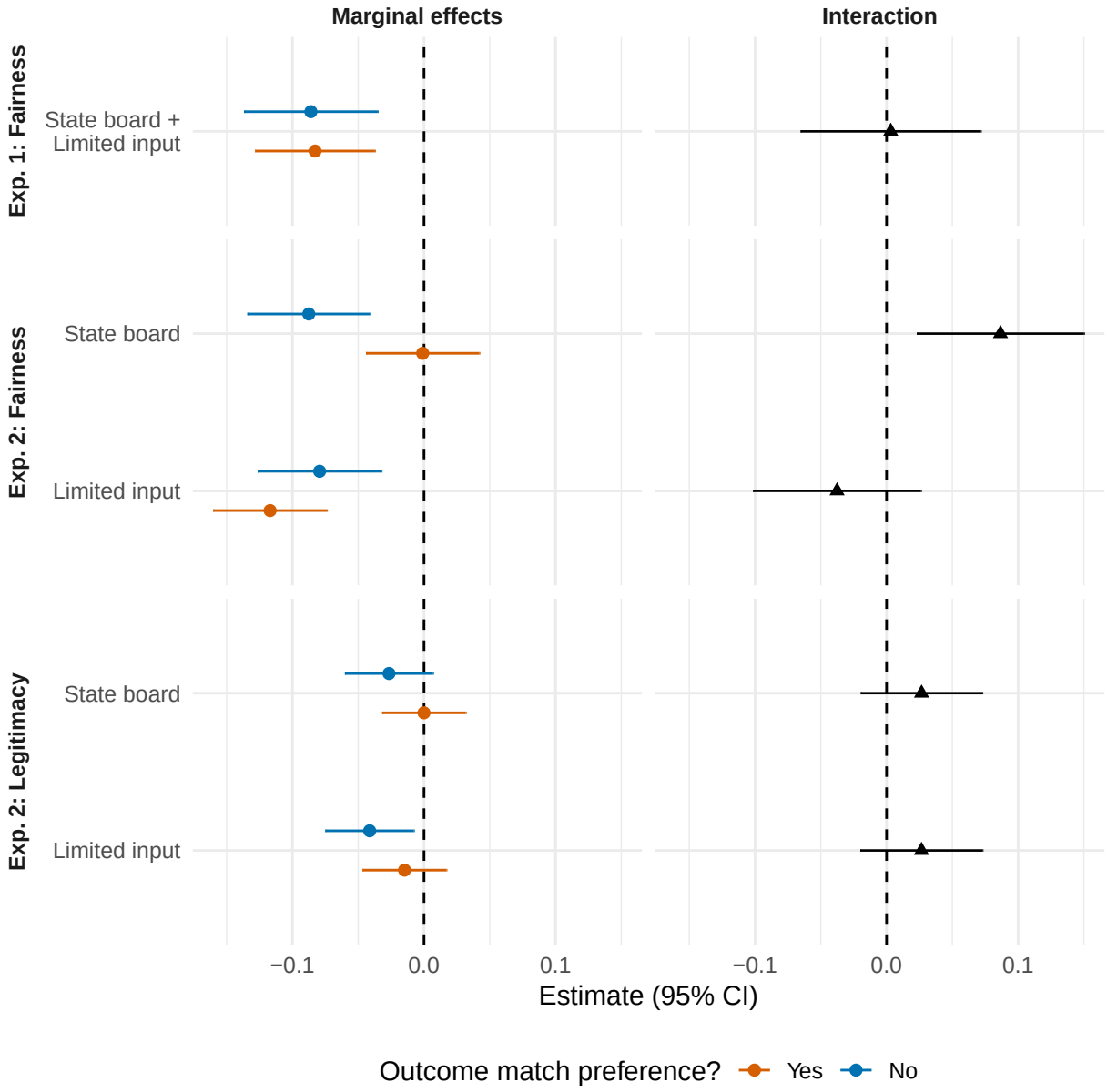

Figure G-9: Interaction between receiving preferred policy outcome and effect of process on perceived fairness and legitimacy of turbine siting decision when the outcome is unknown (Branch 2 only). Point estimates shown with 95% confidence intervals.

H Expected Outcomes

In Experiment 2, respondents in both Branches 1 and 2 stated their expectations of whether the proposed wind far would be approved or denied following the description of the approval process. Thus, I use both branches in evaluating the effect of process on expectations of outcome. There is little evidence that the process systematically affects expectations of approval (Figure H-10).

However, Figure H-11 displays these effects based on respondents' pretreatment support for the wind proposal. Surprisingly, respondents differ in their perceptions of how public input will affect the outcome. Pro-wind respondents believe that less public input will decrease the likelihood of approval, whereas anti-wind respondents believe the opposite. Perhaps both groups believe their neighbors share their attitudes, meaning more public input will secure their preferred outcome.

To be clear, that process affects perceptions of the outcome does not negate the independent effect of process on fairness and legitimacy. Respondents may expect different outcomes based on the process, but these outcomes still do not alter the legitimizing effects of procedural fairness.

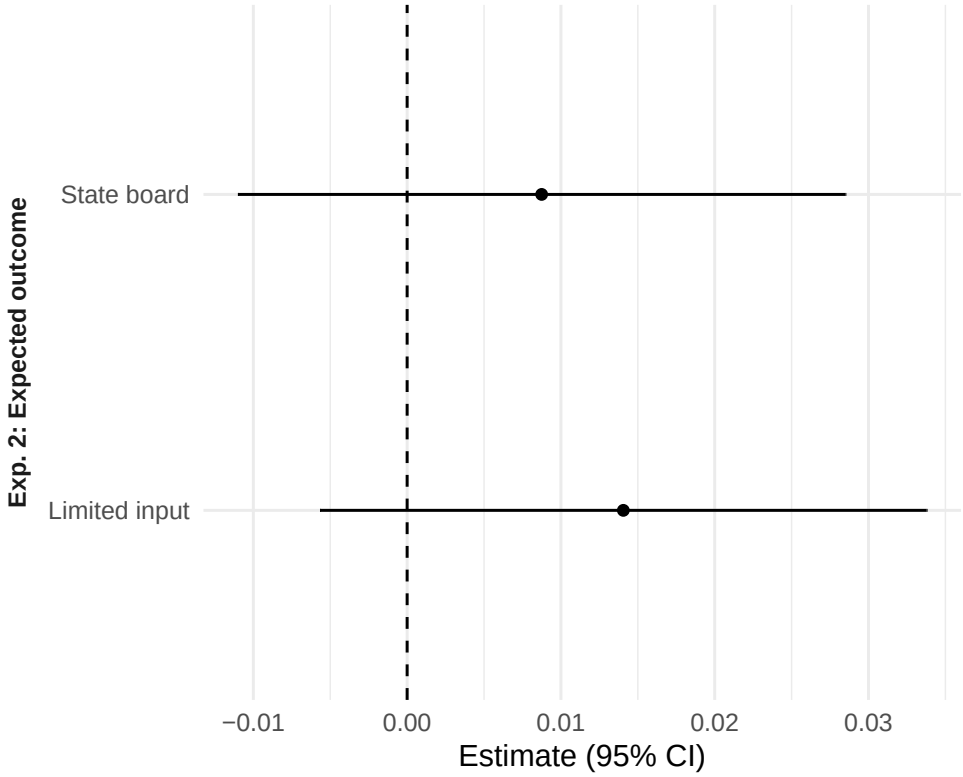

Figure H-10: Effect of process on expected outcome of turbine siting decision when the outcome is unknown (Branches 1 and 2). Point estimates shown with 95% confidence intervals.

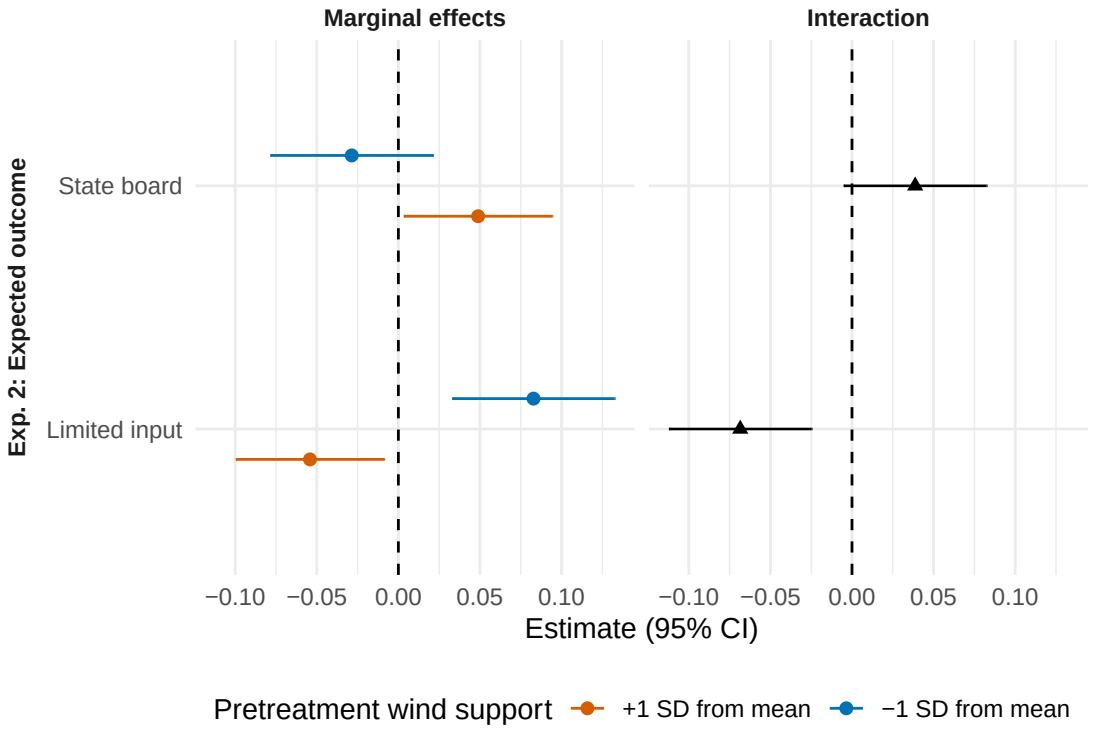

Figure H-11: Effect of process on expected outcome of turbine siting decision when the outcome is unknown (Branches 1 and 2) by respondents pretreatment support for the wind farm. Point estimates shown with 95% confidence intervals.

I Results in Tabular Form

I.1 Effect of Process

	Experiment 1: Fairness	
	Model 1	Model 2
State board, limited input	-0.103^{***}	-0.104^{***}
	(0.017)	(0.017)
Homeowner		-0.005
		(0.020)
White, non-Hispanic		0.018
		(0.020)
Income (cont.)		0.006
		(0.003)
Female (bi.)		-0.003
		(0.017)
College or more		0.041^{*}
		(0.019)
Age		-0.002^{*}
		(0.001)
Republican (-1, 1)		-0.046^{***}
		(0.010)
(Intercept)	0.546^{***}	0.550^{***}
	(0.012)	(0.035)
R^2	0.035	0.081
Adj. R^2	0.034	0.073
Num. obs.	964	956
RMSE	0.270	0.265

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table I-4: Effect of process on perceived fairness of siting decision (Experiment 1, Branch 1 only). Displayed in Figure 2.

	Experiment 2: Fairness		Experiment 2: Legitimacy	
	Model 1	Model 2	Model 3	Model 4
Limited input	-0.086^{***}	-0.088^{***}	-0.028^{**}	-0.030^{**}
	(0.015)	(0.015)	(0.010)	(0.010)
State board	-0.032^{*}	-0.031^{*}	-0.024^{*}	-0.024^{*}
	(0.015)	(0.015)	(0.010)	(0.010)
Homeowner		0.005		−0.021
		(0.018)		(0.012)
White, non-Hispanic		0.009		0.022
		(0.021)		(0.014)
Income (cont.)		−0.006		−0.001
		(0.003)		(0.002)
Female (bi.)		−0.022		−0.004
		(0.015)		(0.010)
College or more		0.010		0.013
		(0.018)		(0.012)
Age		-0.002^{***}		-0.002^{***}
		(0.000)		(0.000)
Republican (-1, 1)		-0.036^{***}		-0.025^{***}
		(0.008)		(0.006)
(Intercept)	0.602^{***}	0.746^{***}	0.524^{***}	0.631^{***}
	(0.012)	(0.030)	(0.008)	(0.020)
R^2	0.024	0.056	0.009	0.055
Adj. R^2	0.023	0.051	0.008	0.050
Num. obs.	1564	1527	1536	1500
RMSE	0.287	0.284	0.194	0.191

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table I-5: Effect of process on perceived fairness and legitimacy of siting decision (Experiment 2, Branch 1 only). Displayed in Figure 2.

	Model 1	Model 2	Model 3	Model 4
State board	-0.104^{***}	-0.091^{***}	-0.103^{***}	-0.095^{***}
	(0.017)	(0.016)	(0.017)	(0.017)
Homeowner	-0.005	0.004	-0.003	0.002
	(0.020)	(0.019)	(0.020)	(0.020)
White, non-Hispanic	0.018	0.009	0.017	0.031
	(0.020)	(0.020)	(0.020)	(0.020)
Income (cont.)	0.006	0.006	0.006	0.006
	(0.003)	(0.003)	(0.003)	(0.003)
Female (bi.)	-0.003	0.013	-0.008	-0.001
	(0.017)	(0.016)	(0.017)	(0.017)
College or more	0.041^{*}	0.038^{*}	0.039^{*}	0.041^{*}
	(0.019)	(0.018)	(0.018)	(0.018)
Age	-0.002^{*}	-0.001	-0.001^{*}	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Republican (-1, 1)	-0.046^{***}	-0.001	-0.019	-0.030^{**}
	(0.010)	(0.010)	(0.012)	(0.010)
Pretreatment project support		0.328^{***}		
		(0.031)		
Climate change concern			0.162^{***}	
			(0.044)	
Expected effects on property				0.309^{***}
				(0.045)
(Intercept)	0.550^{***}	0.312^{***}	0.418^{***}	0.369^{***}
	(0.035)	(0.040)	(0.051)	(0.043)
R^2	0.081	0.189	0.095	0.137
Adj. R^2	0.073	0.181	0.086	0.129
Num. obs.	956	956	956	955
RMSE	0.265	0.249	0.263	0.257

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table I-6: Effect of state board with limited input on perceived fairness of siting decision, alternative specification (Experiment 1, Branch 1 only). Not displayed, but alluded to in main text.

	Experiment 1: Fairness	
	Model 1	Model 2
State board, limited input	-0.096^{***}	-0.097^{***}
	(0.017)	(0.017)
Prefer local government	0.171^{***}	0.167^{***}
	(0.022)	(0.023)
State board, limited input x Local preference	-0.057	-0.063
	(0.034)	(0.033)
Homeowner		-0.011
		(0.019)
White, non-Hispanic		0.025
		(0.019)
Income (cont.)		0.003
		(0.003)
Female (bi.)		-0.002
		(0.017)
College or more		0.029
		(0.018)
Age		-0.002^{***}
		(0.001)
Republican (-1, 1)		-0.035^{***}
		(0.010)
(Intercept)	0.541^{***}	0.593^{***}
	(0.011)	(0.034)
R^2	0.109	0.144
Adj. R^2	0.106	0.135
Num. obs.	964	956
RMSE	0.260	0.256

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table I-7: Effect of process—interacted with preference for local government—on perceived fairness and legitimacy of siting decision (Experiment 1, Branch 1 only). Displayed in Figure S8.

	Experiment 2: Fairness		Experiment 2: Legitimacy	
	Model 1	Model 2	Model 3	Model 4
Limited input	-0.087^{***}	-0.089^{***}	-0.029^{**}	-0.030^{**}
	(0.014)	(0.015)	(0.010)	(0.010)
State board	-0.030^{*}	-0.029^{*}	-0.023^{*}	-0.023^{*}
	(0.014)	(0.015)	(0.010)	(0.010)
Prefer local government	-0.042	-0.034	-0.012	-0.004
	(0.025)	(0.024)	(0.015)	(0.015)
Limited input x Local preference	0.036	0.025	0.014	0.009
	(0.031)	(0.030)	(0.020)	(0.020)
State board x Local preference	-0.056	-0.056	-0.039	-0.043^{*}
	(0.031)	(0.031)	(0.020)	(0.020)
Homeowner		0.006		-0.021
		(0.018)		(0.012)
White, non-Hispanic		0.009		0.022
		(0.021)		(0.014)
Income (cont.)		-0.006		-0.001
		(0.003)		(0.002)
Female (bi.)		-0.024		-0.005
		(0.015)		(0.010)
College or more		0.012		0.014
		(0.018)		(0.012)
Age		-0.002^{***}		-0.002^{***}
		(0.000)		(0.000)
Republican (-1, 1)		-0.036^{***}		-0.025^{***}
		(0.008)		(0.006)
(Intercept)	0.602^{***}	0.743^{***}	0.524^{***}	0.631^{***}
	(0.012)	(0.030)	(0.008)	(0.020)
R^2	0.035	0.066	0.015	0.061
Adj. R^2	0.032	0.058	0.012	0.054
Num. obs.	1563	1526	1535	1499
RMSE	0.286	0.283	0.194	0.190

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table I-8: Effect of process—interacted with preference for local government—on perceived fairness and legitimacy of siting decision (Experiment 2, Branch 1 only). Displayed in Figure S8.

I.2 Resilience of Process

	Experiment 1: Fairness	
	Model 1	Model 2
State board, limited input	-0.103^{***}	-0.104^{***}
	(0.017)	(0.017)
Outcome known	0.005	0.005
	(0.017)	(0.017)
State board, limited input x Outcome known	0.021	0.020
	(0.024)	(0.024)
Homeowner		0.008
		(0.014)
White, non-Hispanic		0.013
		(0.014)
Income (cont.)		0.002
		(0.002)
Female (bi.)		-0.016
		(0.012)
College or more		0.027^{*}
		(0.013)
Age		-0.002^{***}
		(0.000)
Republican (-1, 1)		-0.044^{***}
		(0.007)
(Intercept)	0.546^{***}	0.576^{***}
	(0.012)	(0.027)
R^2	0.029	0.064
Adj. R^2	0.028	0.060
Num. obs.	1995	1979
RMSE	0.273	0.268

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table I-9: Effect of process—interacted with knowledge of decision outcome—on perceived fairness and legitimacy of siting decision (Experiment 1, Branches 1 and 2). Displayed in Figure 3.

	Experiment 2: Fairness		Experiment 2: Legitimacy	
	Model 1	Model 2	Model 3	Model 4
Limited input	-0.086^{***}	-0.088^{***}	-0.028^{**}	-0.029^{**}
	(0.015)	(0.015)	(0.010)	(0.010)
State board	-0.032^{*}	-0.033^{*}	-0.024^{*}	-0.024^{*}
	(0.015)	(0.015)	(0.010)	(0.010)
Outcome known	0.020	0.019	0.025^{*}	0.026^{*}
	(0.018)	(0.018)	(0.012)	(0.012)
Limited input x Outcome known	-0.005	-0.006	-0.001	-0.002
	(0.021)	(0.021)	(0.015)	(0.015)
State board x Outcome known	-0.003	-0.002	0.010	0.013
	(0.021)	(0.021)	(0.015)	(0.015)
Homeowner		-0.003		-0.011
		(0.013)		(0.009)
White, non-Hispanic		-0.013		0.002
		(0.016)		(0.010)
Income (cont.)		-0.005		-0.001
		(0.002)		(0.002)
Female (bi.)		-0.010		-0.003
		(0.011)		(0.008)
College or more		0.020		0.019^{*}
		(0.013)		(0.010)
Age		-0.001^{*}		-0.001^{***}
		(0.000)		(0.000)
Republican (-1, 1)		-0.029^{***}		-0.015^{***}
		(0.006)		(0.004)
(Intercept)	0.602^{***}	0.684^{***}	0.524^{***}	0.585^{***}
	(0.012)	(0.024)	(0.008)	(0.016)
R^2	0.025	0.039	0.012	0.026
Adj. R^2	0.024	0.036	0.010	0.022
Num. obs.	3076	3004	3022	2953
RMSE	0.294	0.292	0.204	0.203

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table I-10: Effect of process—interacted with knowledge of decision outcome—on perceived fairness and legitimacy of siting decision (Experiment 2, Branches 1 and 2). Displayed in Figure 3.

	Experiment 1: Fairness	
	Model 1	Model 2
State board, limited input	-0.086^{**}	-0.086^{***}
	(0.026)	(0.026)
Outcome match	0.139^{***}	0.137^{***}
	(0.025)	(0.025)
State board, limited input x Outcome match	0.006	0.003
	(0.035)	(0.035)
Homeowner		0.017
		(0.021)
White, non-Hispanic		0.010
		(0.021)
Income (cont.)		0.001
		(0.003)
Female (bi.)		-0.014
		(0.018)
College or more		0.002
		(0.019)
Age		-0.002^{**}
		(0.001)
Republican (-1, 1)		-0.043^{***}
		(0.010)
(Intercept)	0.483^{***}	0.538^{***}
	(0.019)	(0.041)
R^2	0.085	0.114
Adj. R^2	0.082	0.104
Num. obs.	953	946
RMSE	0.270	0.267

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table I-11: Effect of process—interacted with outcome match—on perceived fairness and legitimacy of siting decision (Experiment 1, Branch 2 only). Displayed in Figure 4 and Figure S9.

	Experiment 2: Fairness		Experiment 2: Legitimacy	
	Model 1	Model 2	Model 3	Model 4
Limited input	-0.079^{***}	-0.079^{***}	-0.039^{*}	-0.041^{*}
	(0.024)	(0.024)	(0.017)	(0.017)
State board	-0.088^{***}	-0.088^{***}	-0.032	-0.027
	(0.024)	(0.024)	(0.017)	(0.017)
Outcome match	0.157^{***}	0.159^{***}	0.114^{***}	0.114^{***}
	(0.027)	(0.027)	(0.020)	(0.020)
Limited input x Outcome match	-0.033	-0.038	0.024	0.027
	(0.032)	(0.033)	(0.023)	(0.024)
State board x Outcome match	0.086^{**}	0.087^{**}	0.031	0.027
	(0.032)	(0.032)	(0.023)	(0.024)
Homeowner		-0.020		-0.002
		(0.022)		(0.015)
White, non-Hispanic		-0.033		-0.014
		(0.026)		(0.018)
Income (cont.)		-0.004		0.000
		(0.004)		(0.003)
Female (bi.)		0.016		0.004
		(0.017)		(0.012)
College or more		0.033		0.025
		(0.020)		(0.015)
Age		0.000		-0.000
		(0.001)		(0.000)
Republican (-1, 1)		-0.023^{*}		-0.008
		(0.009)		(0.007)
(Intercept)	0.558^{***}	0.581^{***}	0.496^{***}	0.501^{***}
	(0.021)	(0.039)	(0.014)	(0.027)
R^2	0.121	0.135	0.110	0.113
Adj. R^2	0.118	0.126	0.107	0.104
Num. obs.	1315	1290	1292	1269
RMSE	0.294	0.292	0.210	0.210

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table I-12: Effect of process—interacted with outcome match—on perceived fairness and legitimacy of siting decision (Experiment 2, Branch 2 only). Displayed in Figure 4 and Figure S9.